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Burning Mouth Syndrome: A Systematic Review of Treatments

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ABSTRACT

Burning mouth syndrome (BMS) is a chronic oral pain syndrome that primarily affects peri- and postmenopausal women. It is characterized by oral mucosal burning and may be associated with dysgeusia, paresthesia, dysesthesia, and xerostomia. The etiology of the disease process is unknown, but is thought to be neuropathic in origin. The goal of this systematic review was to assess the efficacy of the various treatments for BMS. Literature searches were conducted through PubMed, Web of Science, and Cochrane Library databases, which identified 22 randomized, controlled trials. Eight studies examined alpha-lipoic acid (ALA), three clonazepam, three psychotherapy and two capsaicin, which all showed modest evidence of potentially decreasing pain/burning. Gabapentin was seen in one study to work alone and synergistically with ALA. Other treatments included vitamins, benzydamine hydrochloride, bupivacaine, Catuama, olive oil, trazodone, urea and Hypericum perforatum. Of these other treatments, Catuama and bupivacaine were the only ones with significant positive results in symptom improvement. ALA, topical clonazepam, gabapentin, and psychotherapy may provide modest relief of pain in BMS. Gabapentin may also boost the effect of ALA. Capsaicin is limited by its side effects. Catuama showed potential for benefit. Future studies with standardized methodology and outcomes containing more patients are needed.

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INTRODUCTION

Burning mouth syndrome (BMS) is a diagnosis of exclusion characterized by chronic oral mucosal pain or discomfort without an identifiable causative lesion. It affects over 1 million Americans, especially postmenopausal women between 50 and 70 years of age (Thoppay et al, 2013). A variety of symptoms are present in BMS, including burning or itching of oral mucosa that is often accompanied by paresthesia, dysgeusia, and dysesthesia (Maltsman-Tseikhin et al, 2007). Notably, BMS is often comorbid with xerostomia, nutritional deficiencies, and/or psychological disorders such as anxiety and depression (Klasser et al, 2008).

The pathophysiology of BMS is uncertain. However, a neuropathic etiology related to trigeminal dysfunction is favored (Forsell et al, 2002). Anxiety and depression have also been implicated in the development of BMS, by an unknown mechanism (Galli et al, 2016). Diagnosing BMS may be anxiety-provoking even for the physician given its non-specific symptoms, our limited knowledge of its pathogenesis, and most importantly, our lack of an effective, proven treatment regimen. A 2005 Cochrane review of BMS treatment modalities included 9 trials and found that only 3 interventions showed improvement in symptoms: alpha-lipoic acid, clonazepam, and cognitive behavioral therapy (Zakrzewska et al, 2005). It was concluded that no treatment led to a significant reduction in symptoms, which may have been due to study limitations.

There are currently no universally accepted guidelines for managing BMS. Therefore, we sought to systematically review treatments for BMS in the adult population to incorporate research from the past decade, and to provide a basis for the development of more reliable and effective management strategies. The PICO (Patient, Intervention, Control, Outcome) question

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for this study was, “In patients with BMS, which interventions are effective in improving pain and/or relieving associated symptoms compared to no intervention or placebo?”

METHODS

This study was exempt from the Institutional Review Board as only information in the public domain was used. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were followed, and the Institute of Medicine (IOM) standards for systematic reviews were used as a guide (Moher et al, 2009; Eden et al 2011).

Search Strategy

A query of PubMed/MEDLINE from 1/1/1950 to 5/1/2016 was conducted using the MeSH terms “burning mouth syndrome” and “glossalgia” with the subheadings “drug therapy,” “prevention and control,” and “therapy.” The search was restricted to the English language. A search of the Cochrane Library was also performed using the same MeSH terms for the entire database. A search of Web of Science (WoS) was conducted using the terms “burning mouth syndrome” and “glossalgia,” with the results restricted to English language, randomized control trials published between 1/1/1950 to 5/1/2016 listed under the WoS categories “Dentistry Oral Surgery Medicine” and “Otorhinolaryngology.”

Study Selection

Initially, studies were excluded based on assessment of relevance of title and abstract conducted by three authors independently. The remaining studies were evaluated in their entirety by the same authors and those which met the following criteria were included: randomized, placebo-controlled trial (RCT); diagnosis of BMS based on the International Association for the Study of

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Pain (IASP) definition; and change in pain as primary outcome with relief in associated symptoms as secondary outcomes. Disagreements were resolved with input from the senior author (J.I.).

Study Quality Assessment

The same 3 authors who performed the study selection also performed the bias assessment. Using the Cochrane Collaboration tool for assessing risk of bias in RCTs, the following domains were evaluated for each study included in the review: selection bias (e.g. inappropriate randomization), performance bias (e.g. not blinding researchers giving treatment), detection bias (e.g. not blinding researchers assessing outcome), attrition bias (e.g. patient withdrawals from study), and reporting bias (e.g. selective presentation of data) (Higgins et al, 2011). Each domain was assigned either high, low, or unclear risk. Disagreements were resolved with input from the senior author (J.I.).

Statistics

Mean difference versus placebo in pain scores as indicated by the visual analog scale (VAS) and relative risk ratios (RR) for pain and/or burning type symptom improvement were extracted from the relevant studies along with 95% confidence intervals (CI) when possible. For interventions tested in 2 or more studies, a combined mean VAS score or RR is reported with pooled 95% CIs when appropriate. Forest plots were constructed to show these effects. However, formal meta-analyses and network meta-analyses were not deemed suitable for this systematic review due to significant heterogeneity in data (i.e. different VAS scales used with various scoring systems, different time points for assessing outcomes after treatment, different types of treatments and

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different dosing of medications of the same type, different treatment regimens, different types of outcome assessed such as burning pain versus overall pain versus general improvement, and incomplete data with unreported variance).

RESULTS

Of 433 total studies identified during the initial database searches, 22 remained for final review after duplicates were removed and studies were eliminated based on our inclusion criteria. A flow diagram of study selection is shown in FIGURE 1.

Studies Reviewed

As primary outcome measures, the visual analog scale (VAS) or visual numeric scale (VNS, used in 1 study) with a 0-10 scoring system was used in 13 RCTs to measure pain and burning related symptoms in BMS patients (Femiano and Scully, 2002; Cavalcanti and da Silveira, 2009; López-Jornet et al, 2009; Carbone et al, 2009; Palacios-Sánchez et al, 2015; Petruzzi et al, 2004; Silvestre et al, 2012; Tammiala-Salonen and Forssell, 1999; Cano-Carillo et al, 2014; Rodríguez de Rivera Campillo et al, 2010; Spanemberg et al, 2012; Sardella et al, 2008; Treldal et al, 2016). Other VAS scoring systems were used in 3 studies, with scales of 0-3, 1-7, and 0-8 (Femiano et al, 2004; Bergdahl et al, 1995; Sardella et al, 1999). Arbitrary, non-visual scales for assessing pain/burning type symptom improvement were used 4 studies (Femiano et al, 2000; López-D'alessandro and Escovich 2011; Heckmann et al, 2012; Gremeau-Richard et al, 2004). Other outcome measure tools were broad and varied, including the McGill Pain Questionnaire (MPQ), Beck Depression Inventory (BDI), Zerssen Mood Scale (ZMS), oral health impact profile-14 (OHIP-14), short form survey-36 (SF-36), hospital anxiety and depression scale (HADS), quantitative sensory testing (QST), xerostomia questionnaire, orofacial pain questionnaire,

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Sniffin' Sticks, and salivary flow rate (Carbone et al, 2009; Tammiala-Salonen and Forssell, 1999; Cano-Carillo et al, 2014; Heckmann et al, 2012; Silba et al, 2014, Miziara et al, 2009). A summary of each study reviewed is shown in TABLE I. Forest plots were constructed for both mean difference in pain scores versus placebo (FIGURE 2) and for relative risk of improvement in pain/burning type symptoms associated with BMS (FIGURE 3).

Bias Assessment

Assessment of bias according to the domains indicated in Methods is presented in TABLE II. Randomization appeared adequate in most (95%) of studies. In one study, the method of randomization was not clear (Femiano et al, 2004). Performance bias was low in 85% studies. Two studies involving psychotherapy had high risk for this bias given the difficult of blinding such treatments (Carbone et al, 2009; Bergdahl et al, 1995). Detection bias results followed that of performance bias as blinding usually occurred throughout the studies, and appeared adequate in 80%. Twenty percent of studies suffered from attrition bias. There was a high rate of possible reporting bias (40-60%), as many studies reported either mean pain score changes or improvement, but not both. Also, statistical analyses were not performed to the full extent in several studies to answer many questions that arise from the data. There was an overall lack of uniformity in study methodology, which makes data pooling difficult and dampens the ability to perform meta-analyses on efficacy of treatments.

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Treatment Modalities

Alpha-Lipoic Acid

ALA was the most common therapy studied (in 8 studies), including 1 study which examined the effect of ALA with and without gabapentin, 1 of ALA with and without cognitive psychotherapy (CPT), and 1 of ALA with and without vitamins (Carbone et al, 2009; Femiano et al, 2004; López-D'alessandro and Escovich, 2011). The regimen generally consisted of 200-800mg of ALA per day for 1-2 months. Of those studies which compared change in mean pain scores between ALA and placebo (3 studies), none showed a significant difference (Cavalcanti and de Silveira, 2009; López-Jornet et al, 2009; Carbone et al, 2009). However, when looking simply at improvement, 5 of 6 studies showed a significant difference between ALA and placebo (Femiano and Scully 2002; Palacios-Sánchez et al, 2015; Femiano et al, 2004; Femiano et al, 2000; López-D'alessandro and Escovich, 2011). Both studies examining ALA combined with another treatment (gabapentin or CPT) also reported significant symptom improvement greater than ALA alone (Femiano et al, 2004; López-D'alessandro and Escovich, 2011). One crossover study of 31 patients reported headache and gastric upset as the most common complaints associated with ALA, but the rates of occurrence were not significantly different from placebo (Cavalcanti and de Silveira, 2009).

Clonazepam

Two studies on topical clonazepam (holding tablet in mouth for 3 minutes) demonstrated both significantly decreased pain scores and improvement of pain/burning symptoms versus placebo in BMS patients (Rodríguez de Rivera Campillo et al, 2010; Gremeau-Richard et al, 2004). A third study of ingested clonazepam also reported a significant decrease in pain scores greater

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than placebo, but reanalysis of the data presented failed to show a significant change from when therapy was initiated to when it was stopped (Heckmann et al, 2012). The study also failed to show a difference in change in taste, smell, and salivary flow rate versus placebo. Interestingly, 1 study on topical clonazepam prescribed the medication on a PRN basis, up to 4 times a day, which no other study reviewed in treatment of BMS had done (Rodríguez de Rivera Campillo et al, 2010). This study gave detailed data and analysis of all 66 patients, including a demonstration of the gradual increase in number of pills used in the placebo group over time versus treatment group, and showed a convincing, significant benefit to the medication. Side effects included drowsiness, increased oral burning, and xerostomia, but there was no difference in side effect profile between treatment and control groups (Gremeau-Richard et al, 2004).

Gabapentin

Only one study which used gabapentin met inclusion criteria (López-D'alessandro and Escovich, 2011). In this crossover, placebo-controlled trial, 50% of patients on Gabapentin displayed improved pain scores, compared to only 15% in the placebo arm. In this same trial, gabapentin also appeared to work synergistically with ALA when taken simultaneously to improve pain in 70% of patients, versus 15% in placebo group.

Capsaicin

One study reported significant improvement in pain scores versus placebo using capsaicin oral rinse (Silvestre et al, 2012). Another reported significant reduction of burning symptoms versus placebo with ingested capsaicin capsules (Petruzzi et al, 2004). Notably, topical capsaicin actually intensified the burning sensation in a third of the patients for up to 20 minutes after

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using the rinse (Silvestre et al, 2012). Furthermore, ingested capsaicin produced increasing numbers, up to 32% of the study group, of patients who had gastric pain as treatment time progressed from 1 to 4 weeks (Petruzzi et al, 2004).

Psychotherapy

Psychotherapy (PT) alone versus placebo was examined in 2 studies. One showed a significant decrease in pain versus placebo in patients undergoing cognitive therapy (CT) (Bergdahl et al, 1995). The other demonstrated group PT improved pain symptoms on the MPQ in a significantly greater number of patients than placebo (Miziara et al, 2009). In general PT was used for 3 months or longer, except in the case where it was used in combination with ALA to produce a greater effect (Femiano et al, 2004).

Other Treatments

Benzydamine hydrochloride rinse, lycopene-enriched extra virgin olive oil, trazodone, topical urea, and hypericum perforatum all failed to achieve significant pain or symptom reduction versus placebo (Tammiala-Salonen and Forssell, 1999; Cano-Carillo et al, 2004; Sardella et al, 1999; Silva et al, 2014, Sardella et al, 2008). However, Catuama, a Brazilian herbal product consisting of extracts from the medicinal plants guarana, catuaba, ginger, and muirapuama, did significantly decrease pain versus placebo after an 8 week treatment cycle in 30 patients (Spanemberg et al, 2012). Bupivacaine lozenges taken 3 times a day also statistically significantly decreased oral burning pain, but the effect was minimal and clinical relevance was likely negligible (Treldal et al. 2016).

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DISCUSSION

The pathophysiology of burning mouth syndrome is poorly understood and widely debated, but is believed to be primarily neuropathic in origin. Peripheral small fiber neuropathy has been supported by superficial biopsy and immunohistochemical staining of markers for pathological changes of the tongue mucosa. Several studies have demonstrated a significantly lower density of epithelial nerve fibers along with evidence of diffuse axonal degeneration of the epithelial and sub-papillary nerve fibers (Lauria et al, 2005; Yilmaz et al, 2007). Subclinical major trigeminal neuropathy is thought to be another possible cause. Taste from the chorda tympani of CN VII and sensation from the lingual nerve of CN V are thought to be in delicate balance (Bartoshuk et al, 2005; Grushka et al, 2003).Chorda tympani hypofunction is thought to disinhibit the trigeminal nerve, leading to lingual nerve hyperfunction and subsequent burning pain, dysgeusia, and xerostomia (Eliav et al, 2007).

More recent pain studies with BMS patients have documented alterations in brain activation patterns in the anterior cingulate cortex, bilateral precuneus, and thalamus in response to pain, patterns that are similar to those found in patients with other neuropathies (Kolkka-Palomaa et al, 2015; Albuquerque et al, 2006; Baliki et al, 2014). Other studies indicate that striatal dopamine may play a role in CNS pain modulation, with low dopamine levels in the putamen being associated with disinhibition of the trigeminal system (Hagelberg et al, 2003; Jääskeläinen, 2012). This seems consistent with the fact that low dopamine levels are found in psychiatric disorders such as depression and anxiety, which have been found to be common comorbidities in BMS patients (Lambert et al, 2000; Moraga-Amara et al, 2014). Overall, it is unclear whether there is a single mechanism that predominates or if multiple processes contribute to the neuropathic pain associated with BMS, making it difficult to treat.

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Therefore, given the lack of guidelines for management of BMS, we sought to assess the treatment options suggested by various researchers through a review of the strongest recent evidence in the literature. Unfortunately, no treatment appears to deliver unequivocal results or anything close to a cure. That being said, because the pathogenesis of BMS is not fully understood, we can only treat its symptoms at this time.

Although ALA was the most frequently studied, results were underwhelming. When looking at percent of patients with improved symptoms, there's support from several studies of a significant benefit (Femiano and Scully, 2002; Palacios-Sánchez et al, 2015; Femiano et al, 2004; Femiano et al, 2000; López-D'alessandro and Escovich, 2011). However, when looking at actual pain score changes, there's a lack of evidence for significant improvement (Cavalcanti et al, 2009; López-Jornet et al, 2009; Carbone et al, 2009). Perhaps this suggests that there may be improvement but not enough in these cohorts to be detectable on a VAS, a validated research pain scale with known difficulties detecting small differences in pain. Given the low number of patients treated, varying from 14 to 48, no solid conclusion can be drawn. The upside is that there were no significant adverse effects, which makes ALA at least a sensible medication for first line therapy in attempts to control symptoms. Although gabapentin was studied with ALA in only 1 study, results were convincingly positive (López-D'alessandro and Escovich, 2011). Smaller studies which were poorly designed and did not meet criteria for this review have shown mixed results using gabapentin (Heckmann et al, 2006; White et al, 2004). Better designed gabapentin trials in burning mouth cohorts are still needed as this drug is one of the most effectively used first-line treatments for glossopharyngeal neuralgia, trigeminal neuralgia, and neuropathic pain in general (Zakrzewska, 2010; O'Connor and Dworkin, 2009).

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Clonazepam was studied less frequently than ALA, but the medication arguably provided a more consistent benefit. In both studies of topical clonazepam, there was both a significant decrease in pain score and more patients with improvement than placebo (Rodríguez de Rivera et al, 2010; Gremeau-Richard et al, 2004). Results from clonazepam treatment must be considered from a nerve excitability suppression perspective, but also from that of a treatment for anxiety. Psychotherapy falls in the same category, with consistent benefits and low adverse effects (Femiano et al, 2004; Bergdahl et al, 1995; Miziara et al, 2009). Capsaicin did show significant benefits in pain improvement, but use of the medication will likely be limited by its adverse effects of temporary pain increase and gastric irritation (Petruzzi et al, 2004; Silvestre et al, 2012). Determining optimal topical solution or cream application strength is important when examining their efficacy. All other therapies were examined essentially in single studies, and beside Catuama, none showed much promise.

Generally, the studies reviewed lacked long term follow up data beyond 2 months. Many studies used the VAS scored 1-10, but some use different scoring systems, and associated symptoms were assessed in various manners. Such methodological differences make comparisons difficult. Furthermore, all studies had fewer than 50 patients in the treatment group, with most having between 10 and 30 subjects. Ideally, future double-blind, randomized, placebo-controlled trials would benefit from the following: a standardized primary outcome measure such as a 1 to 10 point VAS for pain and burning; standardized secondary outcome measures such as the BDI and HADS or other validated surveys for depression and anxiety given the high correlation between BMS and these disorders; a validated quality of life survey; more patients to achieve sufficient power (>80%) to detect at least a 1 or 2 point change in the VAS; follow-up of

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at least 1 year given the chronic nature of BMS symptoms; use of ALA, topical clonazepam, and/or PT as comparison groups in addition to placebo when appropriate.

To achieve the above criteria, physicians may need tools to systemize data gathering in BMS patients as their presentations are generally complex and history taking may be time consuming. We have developed a patient questionnaire (Supplementary Figure 1) based on this literature review, other salient literature, and our experience. This is not a validated survey and its purpose is to aid in gathering standardized information, designing patient-specific treatment strategies, and developing more robust future studies.

Many BMS patients suffer from having to "doctor shop" due to difficulty in diagnosis, and receive inadequate treatment with regard to dosage and timing. Therefore, another equally important aspect of management, especially in BMS, is patient education. Because of the chronic nature of the disease and the underlying psychosocial issues associated with the disorder, we believe (at least anecdotally) that patients tend to do better when they understand their disease and prognosis. Acknowledgment of disease chronicity and symptoms is imperative for patient success. Therefore, we developed a BMS patient information sheet (Supplementary Figure 2) which briefly explains the disorder, progression, treatments, and lists additional resources.

Physician understanding of this chronic disease process, frank and realistic doctor-patient encounters, and patient self-education are vital to BMS management. We have found that having a patient questionnaire saves time during the patient interview, and providing patients with an education form empowers them in the management of their own illness. These tools allow for more realistic expectations from consultations. We are currently designing a BMS step-wise treatment algorithm. Although any treatment algorithm for BMS will likely undergo changes as more evidence for current and new treatments is revealed, we encourage clinicians to use the

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results of this study, our patient questionnaire, and our BMS patient information sheet as tailorable guides for individual practices and future research.

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FIGURE 1. Flowchart of study selection process for systematic review.

FIGURE 2. Difference in mean pain scores versus placebo based on VAS from studies reviewed. Diamonds indicate pooled data based on therapy type.

Legend for FIGURE 2

ALA, alpha-lipoic acid; n/a, not available; vs, versus.

FIGURE 3. Relative risk ratio for improvement in pain or burning symptoms associated with BMS. n/N indicates number improved over total number of patients. Diamonds indicated pooled data based on therapy type.

Legend for FIGURE 3

RR, relative risk ratio; ALA, alpha-lipoic acid; CPT, cognitive psychotherapy.

SUPPLEMENTARY FIGURE 1. Sample BMS screening patient questionnaire

SUPPLEMENTARY FIGURE 2. Sample patient information sheet.

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TABLE I. Summary of studies reviewed sorted by year and author.

Author	Year	Treatment	Regimen	Treatment n	Control n	Total n	Outcome(s)	Results summary
Bergdahl et al.	1995	CT	1 h/wk for 12-15 sessions	15	15	30	VAS burning score (1-7) at 12-15 wk	Δ in pain score 2.8 vs placebo -0.3, significant
Sardella et al.	1999	Benz HCl	0.15% oral rinse TID for 4 wk; no tx for 4 wk	10 each	10	30*	VAS symptom severity score (0-8) at 4 wk	No significant improvement or difference in severity score
Tammiala-Salonen et al.	1999	Trazodone	100mg/d for 4 d then 200mg/d for remaining 8 wk	11	17	28	VAS pain score, MPQ, and BDI at 2, 4, 6, 8 wk	No significant difference in pain reduction
Femiano et al.	2000	ALA	600 mg/d for 20 d, then 200 mg/d for 10 d; placebo for 30 d. Crossover: placebo group treated with test group regimen	42	21	42	Nonspecific symptomology at 30 d	Improvement in 76% (63% in crossover) vs placebo 14%, significant
Femiano and Scully	2002	ALA	200 mg TID for 2 m	30	30	60	Burning symptomology (unspecified VAS) at 2 m and 12 m	Improvement in 97% vs placebo 40% at 2 m, significant; deterioration in symptoms in 27% vs placebo 100% at 12 m, significant
Femiano et al.	2004	ALA, CPT	ALA 600 mg/d for 2 m; CPT two 1 h sessions/wk for 2 m; ALA + CPT	48 each	48	192	VAS nonspecific symptom intensity score at 2 m	Improvement in ALA (81%), CPT alone (40%), ALA + CPT (90%) vs placebo (13%), significant
Gremeau-Richard et al.	2004	Topical clonazepam	1 mg in mouth for 3 min TID for 14 d	22	23	45	Pain intensity score (0-10) at 14 d; % improved at least 50% in pain score	Δ in pain score 2.4 vs placebo 0.6, significant; improvement in 50% vs placebo 13%, significant
Petruzzi et al.	2004	Capsaicin	0.25% capsules TID for 1 m	25	25	50	Unspecified VAS score at 1 m	Improvement in 76% vs 4% placebo, significant
Carbone et al.	2008	ALA, vit	ALA 400 mg BID for 8 wk; ALA 400 mg BID + vit for 8 wk	14 ALA, 18 ALA+vit	20	52	VAS pain score and MPQ at 8 wk and 16 wk	No significant improvement in any tx group
Sardella et al.	2008	Hypericum perforatum extract	300mg TID for 12 wk	19	20	39	VAS oral pain score at 0d, 28d, 56d, and 84d; number of oral mucosa sites with reported burning symptoms	No significant difference in VAS scores. Significantly fewer oral sites involved in test group.

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Cavalcanti and da Silveira	2009	ALA	200 mg TID for 30 d, then 20 d washout period, then placebo for 30 d; crossover: tx and placebo groups switch and repeat cycle	31	31	31	VAS burning score and nonspecific symptomology at 30 d, 50d, 80d	No significant improvement
Lopez-Jornet et al.	2009	ALA	800 mg/d for 8 wk	23	16	39	VAS burning score at 1 m and 2 m	No significant improvement
Miziara et al.	2009	Group PT	1 session/wk for 3 m	24	20	44	MPQ at 3 m	Improvement in 70.8% vs placebo 40%; significant
Rodriguez et al.	2010	Topical clonazepam	0.5 mg tablet in mouth for 3 min then spit out PRN (4 tablets max/d) for 6 m	33	33	66	VAS burning pain score at 1 m and 6 m; % improved at least 50% in pain score	Δ in pain score 4.8 vs placebo 3.3, significant; improvement in 70% vs placebo 13%, significant
Lopez and Escovich	2011	ALA, gaba	ALA 600 mg/d for 2 m; gaba 300 mg/d for 2 m; ALA + gaba	20 each	60	120	Burning at 2 m	Improvement in ALA 55%, GABA 50%, both 70%, placebo 15%, significant
Heckmann et al.	2012	Clonazepam	0.5 mg/d for 9 wk	10	10	20	Pain score, taste test strip, Sniffin' Sticks smell test, salivary flow rate, BDI, and ZMS at 9 wk	Δ in pain score 2.9 vs placebo 1.5, significant; no significant change in taste, salivary flow
Silvestre et al.	2012	Capsaicin	0.02% rinse TID for 1 wk, then 1 wk washout period. Crossover: tx and placebo groups switch	23	23	23	VAS discomfort score over 1 wk measured in the AM and in the PM daily.	Significant decrease in pain score 2.31 in the AM and 2.96 in the PM vs placebo (Δ not reported)
Spanemberg et al.	2012	Catuama	310 mg BID for 8 wk	30	30	60	VNS (0-10) and FS (0-5) at 4, 8, 12 wk	Δ in pain score 3.5 vs placebo 1.4, significant
Silva et al.	2013	Urea	10% oral topical 3-4 times/d for 3 m	19	19	38	Orofacial pain questionnaire, xerostomia questionnaire, QST at 3 m	No significant differences in salivary flow or gustative, olfactory, or sensory thresholds
Cano-Carillo et al.	2014	Olive oil	Lycopene-enriched (300ppm) extra virgin olive oil 1.5 ml spray TID for 12 wk; placebo	26	24	50	VAS symptoms score, OHIP-14, SF-36, and HADS at 12 wk	No significant differences between the groups
Palacios-Sanchez et al.	2015	ALA	200 mg TID for 2 m	25	29	54	VAS symptom score at 2 m	Improvement in 64% vs placebo 27.6%; significant

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Treldal et al.	2016	Bupivacaine lozenge	5mg lozenge TID (after meals) until dissolved for 2 wk, then washout period for 1 wk. Crossover: tx and placebo switched.	18	18	18	VAS score for oral pain, xerostomia, and taste alteration recorded by pt for 1 out of 3 lozenges/day; questionnaire assessing patient perception/acceptance of lozenge formulation	Significant decrease in oral burning pain. No significant effect on xerostomia or taste alteration.

Legend for TABLE I

ALA, alpha-lipoic acid; d, day(s); vs, versus; TID, 3 times a day; m, month(s); VAS, visual analog scale; vit, vitamins; BID, twice a day; wk, week(s); MPQ, McGill Pain Questionnaire; tx, treatment; CPT, cognitive psychotherapy; h, hour; gaba, gabapentin; Δ, change; PRN, as needed; max, maximum; BDI, Beck Depression Inventory; ZMS, Zerssen Mood Scale; CT, cognitive therapy; PT, psychotherapy; Benz HCl, benzydamine hydrochloride; VNS, visual numeric scale; FS, face scale; ppm, parts per million; OHIP-14, oral health impact profile-14; SF-36, short form survey-36; HADS, hospital anxiety and depression scale; QST, quantitative sensory testing.

*Study included a 3rd group of 10 that did not receive any treatment at all. There were no statistically significant differences among the 3 regimens.

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TABLE II. Risk of bias in reviewed studies.

Author	Year	Bias					
		Intervention	Selection	Performance	Detection	Attrition	Reporting
Bergdahl et al.	1995	CT	L	H	H	L	U
Sardella et al.	1999	Benz HCl	L	L	L	L	H
Tammiala-Salonen et al.	1999	Trazodone	L	L	L	H	L
Femiano et al.	2000	ALA	L	U	U	L	H
Femiano and Scully	2002	ALA	L	L	L	L	H
Femiano et al.	2004	ALA, CPT, both	H	H	H	L	H
Gremeau-Richard et al.	2004	Top Clz	L	L	L	L	L
Petruzzi et al.	2004	Capsaicin PO	L	L	L	L	H
Carbone et al.	2008	ALA, ALA + vit	L	L	L	H	U
Sardella et al.	2008	H. perforatum	L	L	L	L	L
Cavalcanti and da Silveira	2009	ALA	L	L	L	L	L
Lopez-Jornet et al.	2009	ALA	L	L	L	H	L
Miziara et al.	2009	Group PT	L	L	U	L	U
Rodriguez et al.	2010	Top Clz	L	L	L	L	L
Lopez and Escovich	2011	ALA, gaba, both	L	L	L	L	L
Heckmann et al.	2012	Clonezapam	L	L	L	L	H
Silvestre et al.	2012	Capsaicin PO	L	L	L	H	H
Spanemberg et al.	2012	Catuama	L	L	L	L	L
Silva et al.	2013	Urea	L	L	L	L	L
Cano-Carrillo et al.	2014	Olive oil	L	L	L	L	U
Palacios-Sanchez et al.	2015	ALA	L	L	L	L	H
Treldal et al.	2016	BUP lozenge	L	L	L	H	L

Legend for TABLE II

ALA, alpha-lipoic acid; U, unclear; H, high; L, low; gaba, gabapentin; Benz HCl, benzydamine hydrochloride; CT, cognitive therapy; Top Clz, topical clonazepam; BUP lozenge, bupivacaine lozenge; H. perforatum, Hypericum performatum

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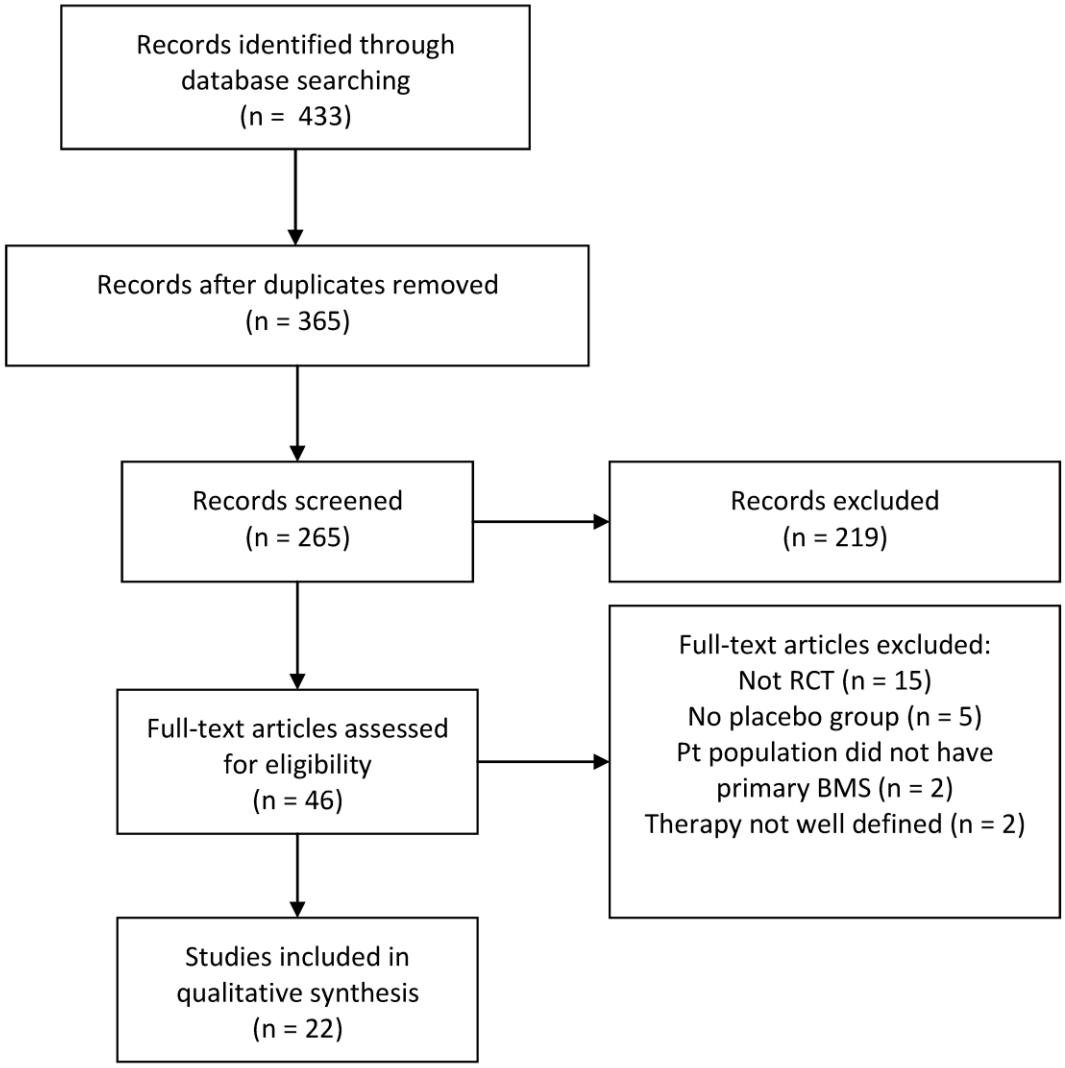

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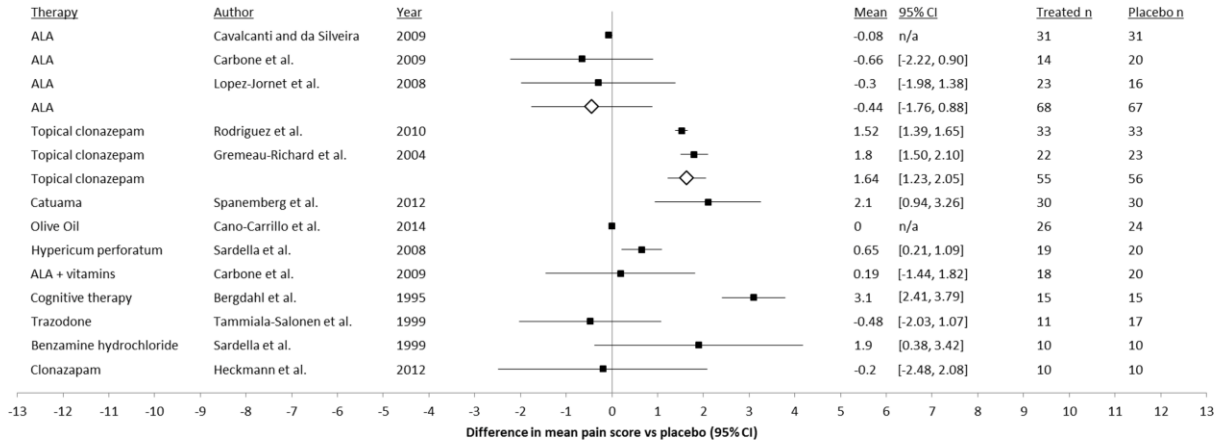

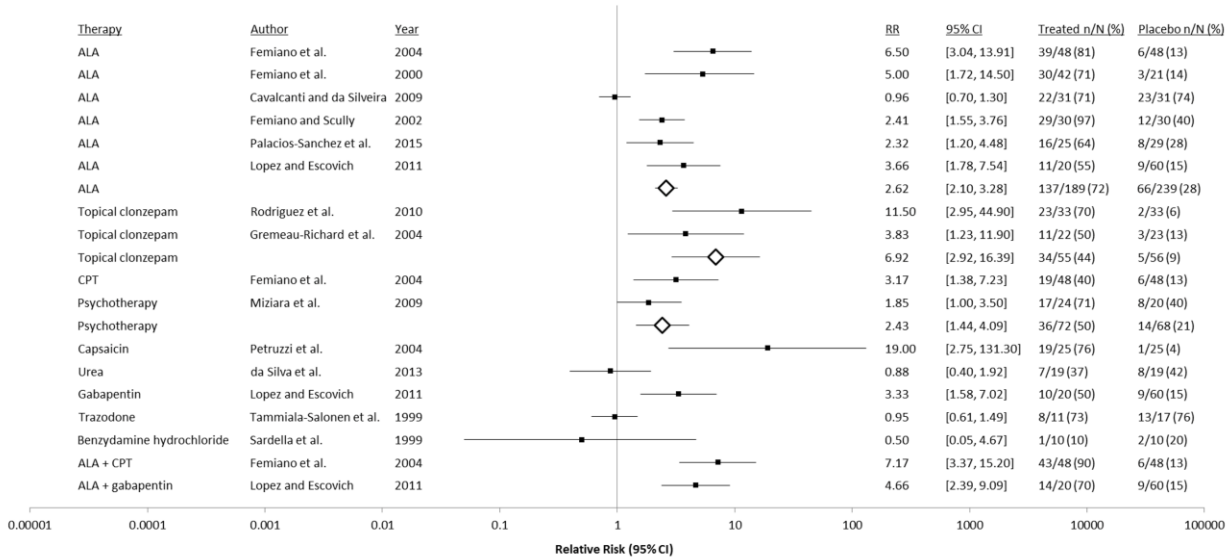

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